

Unveiling causal regulatory mechanisms through cell-state parallax

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Genome-wide association studies (GWAS) identify numerous disease-linked genetic variants at noncoding genomic loci, yet therapeutic progress is hampered by the challenge of deciphering the regulatory roles of these loci in tissue-specific contexts. Single-cell multimodal assays that simultaneously profile chromatin accessibility and gene expression could predict tissue-specific causal links between noncoding loci and the genes they affect. However, current computational strategies either neglect the causal relationship between chromatin accessibility and transcription or lack variant-level precision, aggregating data across genomic ranges due to data sparsity. To address this, we introduce GrID-Net, a graph neural network approach that generalizes Granger causal inference to detect new causal locus–gene associations in graph-structured systems such as single-cell trajectories. Inspired by the principles of optical parallax, which reveals object depth from static snapshots, we hypothesize that causal mechanisms could be inferred from static single-cell snapshots by exploiting the time lag between epigenetic and transcriptional cell states, a concept we term “cell-state parallax.” Applying GrID-Net to schizophrenia (SCZ) genetic variants, we increase variant coverage by 36% and uncovered noncoding mechanisms that dysregulate 132 genes, including key potassium transporters such as *KCNQ2* and *SLC12A6*. Furthermore, we discover evidence for the prominent role of neural transcription-factor binding disruptions in SCZ etiology. Our work not only provides a strategy for elucidating the tissue-specific impact of noncoding variants but also underscores the breakthrough potential of cell-state parallax in single-cell multiomics for discovering tissue-specific gene regulatory mechanisms.

Dynamic changes in the chromatin states of cis-regulatory genomic elements influence gene expression in many biological processes, including development, differentiation, reprogramming, and perturbation response^{1–3}. Correspondingly, genome-wide association studies (GWAS) have implicated thousands of noncoding loci in diseases and

complex traits^{4,5}. Yet, the challenge of linking these noncoding loci to the genes they regulate has impeded therapeutic advances. The common approach of simply associating a locus to its nearest gene is inadequate, as it overlooks distal locus–gene interactions. This is a critical limitation, as distal interactions are likely to be important for

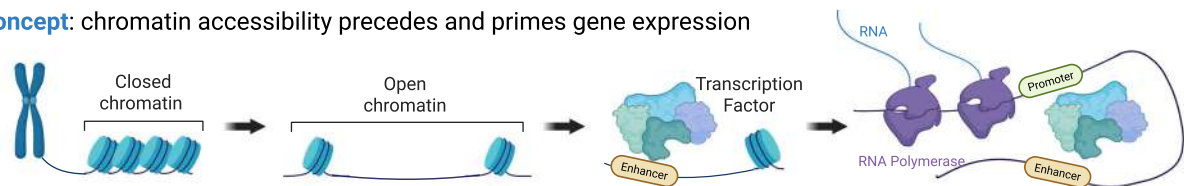
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many genes. Meanwhile, experimental approaches to discover such interactions can be difficult to scale. Expression quantitative trait locus (eQTL) analyses require sample collections from large cohorts⁶, and CRISPR perturbations have been limited to specific loci⁷. Computational methods based on bulk data have also been proposed, such as Wang et al.'s integration of multiple data types to link variants to genes⁸, and Fulco et al.'s activity-by-contact (ABC) model^{9,10} that predicts enhancers of a gene based on the activity and 3D chromatin structure around the gene. However, these models focus on the relative spatial positions of enhancers and genes, and are not designed to evaluate whether changes in accessibility at a locus actually influence gene expression.

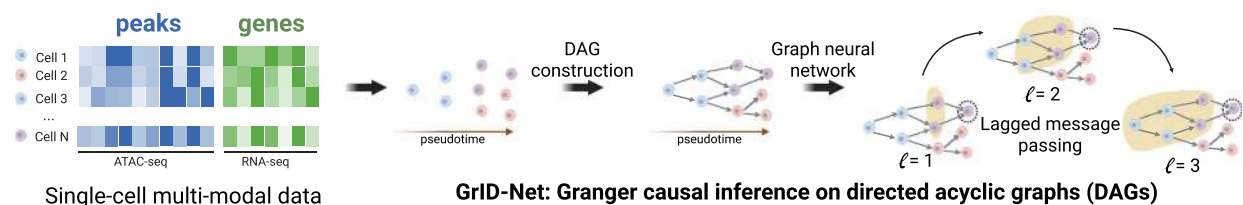
Recent advances in single-cell multimodal technology, which can simultaneously profile chromatin accessibility (scATAC-seq) and gene expression (scRNA-seq) in the same cell, offer a promising opportunity to uncover tissue-specific, genome-scale, causal relationships between noncoding loci and genes in a single experiment^{11–14}. However, current computational methods face substantial hurdles in extracting

meaningful locus-gene associations. Existing approaches have diverse and often unrelated objectives: achieving concordance between the modalities^{15,16}, inferring gene regulatory networks by considering the accessibility of transcription-factor binding motifs^{17,18}, or using chromatin accessibility to improve RNA velocity inference¹⁹. Unfortunately, owing to the extreme sparsity of scATAC-seq data (<1% non-zero entries per cell²⁰), these approaches aggregate chromatin peaks within a defined range of a gene; for instance, MultiVelo consolidates chromatin signals into a single, per-gene score¹⁹. This results in a loss of variant- and locus-level resolution and also risks ignoring distal peaks. Moreover, these methods are largely reliant on correlational analyses between scATAC-seq and scRNA-seq, thereby overlooking the inherent causal time lag in gene regulation (Fig. 1a). For instance, poised enhancers, which play essential roles in development, undergo changes in chromatin states and histone modifications *prior* to activating their target genes and would be missed by a correlational analysis²¹. Consequently, the appropriate epigenetic and transcriptional readouts to be cross-referenced may correspond to subtly different cell states.

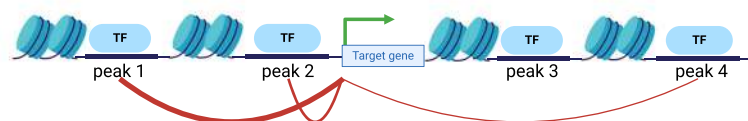
a Concept: chromatin accessibility precedes and primes gene expression



b Analysis: estimate causal peak-gene relationships from single-cell snapshot of a dynamic biological process



c Inference: tissue-specific, genome-wide enhancer-gene associations



d Application: GWAS interpretation and gene prioritization

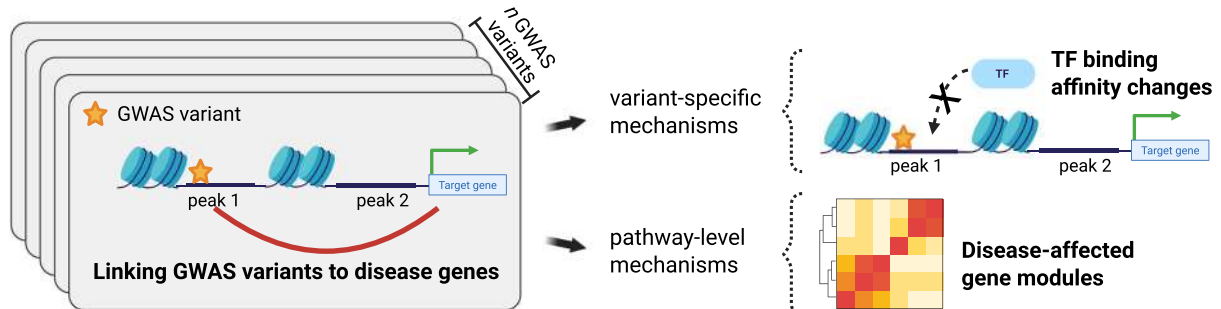


Fig. 1 | GrID-Net Framework. **a** The causal asynchrony between chromatin accessibility and gene expression results in “cell-state parallax”, where a temporal lag exists between these two modalities. Leveraging this parallax from single-cell snapshots enables inference of causal locus-gene associations. **b** To identify links between noncoding loci (i.e., peaks) and the genes they regulate, we apply GrID-Net to single-cell multimodal data in which chromatin accessibility (ATAC-seq) and gene expression (RNA-seq) have been co-assayed in the same cell. GrID-Net is an extension of classical, total-order based, Granger causality to that which can operate on DAG-based representations of single-cell trajectories. The core of GrID-

Net is a graph neural network-based lagged message passing architecture. **c** GrID-Net enables the detection of asynchronous interactions between regulatory activity at accessible noncoding loci and their downstream effects on gene expression. **d** GrID-Net peak-gene links enable the functional interpretation of genetic variants associated with disease. We present a framework for studying the functional effects of such variants both at the level of specific variants and with respect to disease-affected modules. Created in BioRender. Singh, R. (2025) <https://BioRender.com/yfu523x>.

We refer to this temporal discrepancy as “cell-state parallax,” drawing inspiration from the optical concept of parallax, which describes how different vantage points can unveil new insights. Cell phone cameras with dual lenses leverage optical parallax by capturing multiple 2D images of the same object from slightly offset positions to enable 3D depth perception. Analogously, in single-cell multimodal assays, each modality captures a snapshot of the cell’s state, but at an offset due to the temporal lag between the epigenome and the transcriptome. By exploiting this offset, causal relationships between the modalities can be unveiled. However, a fundamental challenge in accounting for these offsets is that single-cell experiments destroy cells upon measurement, so the temporal ordering of single-cell measurements is often unclear. Moreover, an inference framework needs to be robust to the sparsity of both scATAC-seq and scRNA-seq data^{22,23}. Some previous methods mitigate sparsity by averaging transcriptionally similar cells into pseudocells¹¹, but this comes at the cost of losing cell-level granularity, which is crucial for discerning subtle offsets between distinct modalities. In this work, we operate at a per-peak and per-cell resolution, addressing sparsity through our machine learning framework.

To resolve single-cell multimodal snapshots into causal regulatory relationships, we leverage mathematical principles that underlie the dynamics between causes and their effects. We focus on Granger causality, a statistical framework that exploits the idea that a cause precedes, and is therefore predictive of, its downstream effect. There, a temporally causal relationship between two time-varying processes can be inferred solely by autoregressive analysis of time-resolved observational data. Granger causality has been a powerful tool in biology, econometrics, and other domains for causal discovery without the need to perform interventions.

However, the classical framework of Granger causality is limited to settings where observations can be ordered along a single time axis. Mathematically, an ordering of observations can be categorized as *total* or *partial*: in the former, any pair of observations can be ordered unambiguously, while for the latter, only a subset of observations can be so ordered. Partial orderings characterize many dynamical systems, including disinformation cascades in social media, financial transaction networks, and parallel computing systems. In single-cell biology, when a differentiation trajectory has multiple branches, cell states across distinct branches are not directly comparable, necessitating a partial ordering of cells. Classical Granger causal analysis, being limited to total orderings, therefore needs to be adapted for inferring causal single-cell dynamics.

Our key mathematical advance is a generalization of Granger causality to partial orderings. We exploit the equivalence between such orderings and directed acyclic graphs (DAGs) and design a graph neural network framework to infer causality from the dynamics. We complement this with the conceptual advance of cell-state parallax from which we infer a DAG-based representation of single-cell multimodal dynamics. We combine these innovations into GrID-Net (Granger Inference on DAGs), a framework for Granger causal inference on DAGs that we apply to single-cell multimodal data for inferring causal noncoding locus-gene relationships (Fig. 1b, c). Leveraging lagged message-passing on a graph neural network, GrID-Net resolves the cell-state parallax between the epigenome and transcriptome. We demonstrate that it outperforms existing state-of-the-art approaches for accurately identifying noncoding locus-gene links across a variety of cell types and single-cell multimodal assays. We then use GrID-Net to link fine-mapped genetic variants associated with schizophrenia (SCZ) to the genes they affect. We uncover noncoding mechanisms of gene dysregulation that prominently feature disruptions of neural transcription-factor binding and synaptic pathways as putative drivers of SCZ disease pathogenesis (Fig. 1d). Our findings demonstrate the power of GrID-Net in uncovering tissue-specific causal mechanisms of disease and development, using data generated from just one sample.

Our work also illustrates the utility of multimodal single-cell studies for investigating the impact of genetic variants on function and phenotype. The code for GrID-Net can be found at <https://github.com/alexw16/gridnet>, and the full list of SCZ variant-gene links can be found in Supplementary Data 1.

Results

Overview of GrID-Net

In general, inferring true causality can be a challenge, as it requires perturbations to confirm cause-and-effect relationships and validations to rule out potential confounders. Granger causality is a powerful statistical concept that can *approximate* causality in time-varying processes solely from observational data. It leans on the principle that a cause precedes its effect^{24,25}. Given two time-dependent variables x (a potential cause) and y (the effect), if the forecast for y can be improved by accounting for the past values of x , then x is said to Granger cause y because changes in x temporally precede those in y .

One of our key contributions is to broaden the set of domains to which Granger causal analyses can be applied. Specifically, we generalize Granger causal inference to dynamical systems that are characterized by partial orderings. To model the asynchrony between cause and effect, we modify standard graph neural networks to allow for lagged message passing. Each node (i.e., a cell state) in the directed acyclic graph accumulates information from its parent nodes (past states), leveraging this lagged information to forecast the present state in the context of the graph-structured dynamics. GrID-Net consists of two forecasting models: a *reduced* model forecasts an effect y using only the past of y , and a *full* model forecasts the effect y using both the past of y and the past of the candidate causal variable x . The strength of the Granger causal link between x and y is evaluated by a one-tailed F-test, which quantifies the improvement in forecasting accuracy of the *full* model over that of the *reduced* model. A large improvement in forecasting accuracy by the full model indicates that the past values of x are predictive of future values of y , thereby suggesting a significant Granger causal relationship between x and y (Methods).

GrID-Net enables the detection of asynchronous interactions between gene regulatory elements along single-cell trajectories. For each single-cell multimodal dataset, a DAG of cell states is constructed from a k -nearest neighbor graph of cells, with edges oriented in accordance with pseudotime computed after a multimodal integration with Schema²⁰. RNA-velocity based edge orientations that capture local dynamics can also be supplied, so long as any resulting cycles are broken. We evaluated the Granger causal strength between each gene and chromatin accessible element (“peak”) within one megabase (Mb) of the gene’s transcription start site, ranking the peak-gene pairs by their Granger causal strengths (Methods).

GrID-Net more accurately identifies noncoding locus-gene links

We evaluated GrID-Net on four single-cell multimodal datasets (scATAC-seq and scRNA-seq) that characterize a range of biological processes, including cell differentiation and drug treatment response^{11,12,26,27}. We refer to these four datasets as the 10x Multiome, SHARE-seq, SNARE-seq, and sci-CAR datasets. Statistics describing the DAGs associated with each dataset can be found in Supplementary Table 1. For each dataset, we sought to validate noncoding locus-gene link predictions against expression quantitative trait (eQTL) loci profiled in matching tissue²⁸ (Methods). eQTLs are an attractive choice for benchmarking such predictions, as they can show associations between genetic variation at a locus and the corresponding changes in a gene’s expression. While using eQTL mapping to identify specific variant-gene associations can be prone to confounding factors such as linkage disequilibrium²⁹, we reasoned that a systematic assessment against the full collection of eQTLs specific to the matching tissue serves as a useful genome-wide proxy for perturbational validation of causal interactions between genomic loci and genes.

For each gene, we intersected all ATAC-seq peaks within 1 Mb of its transcription start site (TSS) with the set of eQTL variants linked to it, associating each peak with the most significant variant overlapping it. We then evaluated the peak–gene pairs prioritized by GrID-Net and two correlation-based methods used in previous work^{11,27}, Pearson correlation and pseudocell correlation. The latter computes correlations on measurements aggregated across multiple cells. In addition, to assess the advantages of GrID-Net's ability to operate on DAGs, we evaluated the performance of traditional, linear Granger causal inference using vector autoregression (Linear Granger) and a state-of-the-art generalized vector autoregressive method for nonlinear Granger causal inference (GVAR)^{24,30}. We also evaluated the ABC model's predictions on bulk data from matching tissues, comparing it against the predictions from the single-cell datasets (Methods).

We evaluated the accuracy of each method in retrieving peak–gene pairs that overlap significant eQTLs ($P < 10^{-10}$). GrID-Net was significantly more accurate than all of the other methods across all datasets (Fig. 2a, Welch's one-tailed t -test $P < 0.01$), with GVAR's performance on the sci-CAR dataset as the only exception. GrID-Net also exhibited 7–22% higher area under the receiver operating characteristic curve (AUROC) compared to the next best method in 3 of the 4 datasets. GrID-Net's ability to outperform these methods was robust to the choice of the eQTL significance threshold (Supplementary Fig. S1). GrID-Net's accuracy was also highly consistent across various hyperparameter choices that influence the construction of the DAG (Supplementary Fig. S2, Methods). GrID-Net demonstrated the smallest improvement in AUROC for the sci-CAR dataset, although all methods performed poorly on this dataset, likely due to the lower data quality of sci-CAR as the earliest single-cell multimodal assay developed. Meanwhile, the correlation-based methods exhibited near-random performance across all datasets, consistently underperforming not only GrID-Net but also the two Granger causal methods and the bulk data-based ABC model. A similar comparison against a recent multiome-specific approach, TRIPOD³¹ demonstrates GrID-Net's superior performance (Supplementary Fig. S3). GrID-Net's ability to substantially outperform widely used correlation-based methods and existing Granger causal approaches points to its potential to unravel much richer regulatory insights from these multimodal assays than has been possible.

To further benchmark GrID-Net using functional indicators of noncoding locus–gene interactions, we examined three-dimensional chromatin contacts between gene promoters and noncoding loci. Compared to the two correlation-based methods and the existing Granger causal methods, we found that GrID-Net more accurately prioritized peak–gene pairs that were in frequent three-dimensional contact (Fig. 2b, Methods). The ABC method uses bulk Hi-C data as an input, so we did not evaluate its agreement with chromatin contacts. GrID-Net also demonstrated significantly higher accuracy in prioritizing the most well-supported subset of peak–gene pairs that were corroborated by both three-dimensional contacts and eQTL data (Supplementary Fig. S4). GrID-Net's ability to consistently outperform existing methods in detecting peak–gene pairs supported by different but complementary functional indicators further highlights its ability to hone in on potential causal interactions between noncoding loci and genes. In the future, the GrID-Net framework could be expanded to also utilize chromatin contacts data when available—as we describe later, incorporating one-dimensional sequence proximity as a signal into GrID-Net improves its results. However, chromatin contact data is currently available only for a limited set of tissues³², and the ability to make regulatory inferences from just a single multimodal assay enables the convenient study of rare cell types, cancer mutations, and drug responses, etc.

To specifically investigate the benefit of representing tree-like structures of complex trajectories with a DAG, we next compared GrID-Net to applications of standard, linear Granger causality on individual

branches of a trajectory that are more linear in nature. We focused this analysis on the SHARE-seq dataset²⁶, which profiles a tree-like trajectory with three separate branches that define the transition of transit-amplifying cells to cortex, medulla, or inner root sheath (IRS) cells²⁶. We applied linear Granger causality to the full trajectory as well as to each of the three distinct branches separately. While all of the linear Granger causality approaches yielded better-than-random performance in predicting eQTLs, GrID-Net showcased significantly higher accuracy than all of these approaches. This result not only demonstrates the limitations of studying locally linear components of a trajectory in isolation but also provides further evidence for GrID-Net's ability to harness the more complex tree-like structures of full trajectories to more fully capture critical regulatory biology.

We also assessed the robustness of the single-cell methods to sparsity in single-cell data. GrID-Net better handled sparse data (Supplementary Fig. S5), yielding higher accuracies than the other evaluated correlation-based methods for peak–gene predictions involving genes with sparse data (non-zero transcripts in fewer than 10% of cells) or sparse peaks (under 1% of cells). Notably, pseudocell correlation failed to consistently outperform standard Pearson correlation on these sparse peak–gene cases, even though the former is explicitly designed to compensate for the sparsity of measurements at the level of individual cells. This may be because pseudocell averaging reduces the effective number of independent observations and thus relinquishes the statistical advantages of single-cell heterogeneity. Altogether, these analyses indicate not only GrID-Net's robustness to the current sparsity of single-cell data but also the potential for its increased effectiveness as the sensitivity of these multimodal assays continues to improve.

GrID-Net reveals how chromatin accessibility primes gene activity

To better understand the biological basis of GrID-Net's superior accuracy, we applied it on the 10x Multiome dataset profiling human corticogenesis. We then honed in on peak–gene pairs where GrID-Net and correlation-based methods disagree. Specifically, peak–gene pairs were ranked according to the difference between their GrID-Net and correlation-based prioritization scores (Methods), with higher ranking pairs being the ones that were more strongly prioritized by GrID-Net. We then investigated the distinguishing biological characteristics of these differentially-prioritized pairs.

One of our motivations behind GrID-Net is the observation that regions of chromatin around lineage-determining genes become accessible prior to gene expression, foreshadowing lineage choice^{21,26}. GrID-Net was designed to resolve this cell-state parallax between chromatin accessibility and gene expression. We assessed GrID-Net's ability to do so by comparing the lags associated with GrID-Net's detected peak–gene pairs to the lags of peak–gene pairs prioritized by correlation-based methods, which inherently focus on synchronous interactions. We first associated each peak and each gene with the pseudotime corresponding to its highest value and used the difference between these pseudotime points as an estimate of the lag associated with each peak–gene pair. For both correlation-based methods, we found that peak–gene pairs differentially prioritized by GrID-Net are marked by significantly longer pseudotime lags between chromatin accessibility and gene expression ($P < 10^{-26}$, Welch's one-tailed t -test, Fig. 2d, Methods). This result supports GrID-Net's ability to account for the asynchrony between noncoding regulatory activity and downstream gene expression control that are expected to characterize causal interactions between the two. GrID-Net identifies these asynchronous relationships for peak–gene pairs that span a broad range of genomic distances. For example, GrID-Net was able to detect the lagged relationship between the expression of *CBWD5*, a gene implicated in human neuronal development³³, and the accessibility of a promoter region roughly 100 bp upstream of its TSS. In addition, GrID-Net

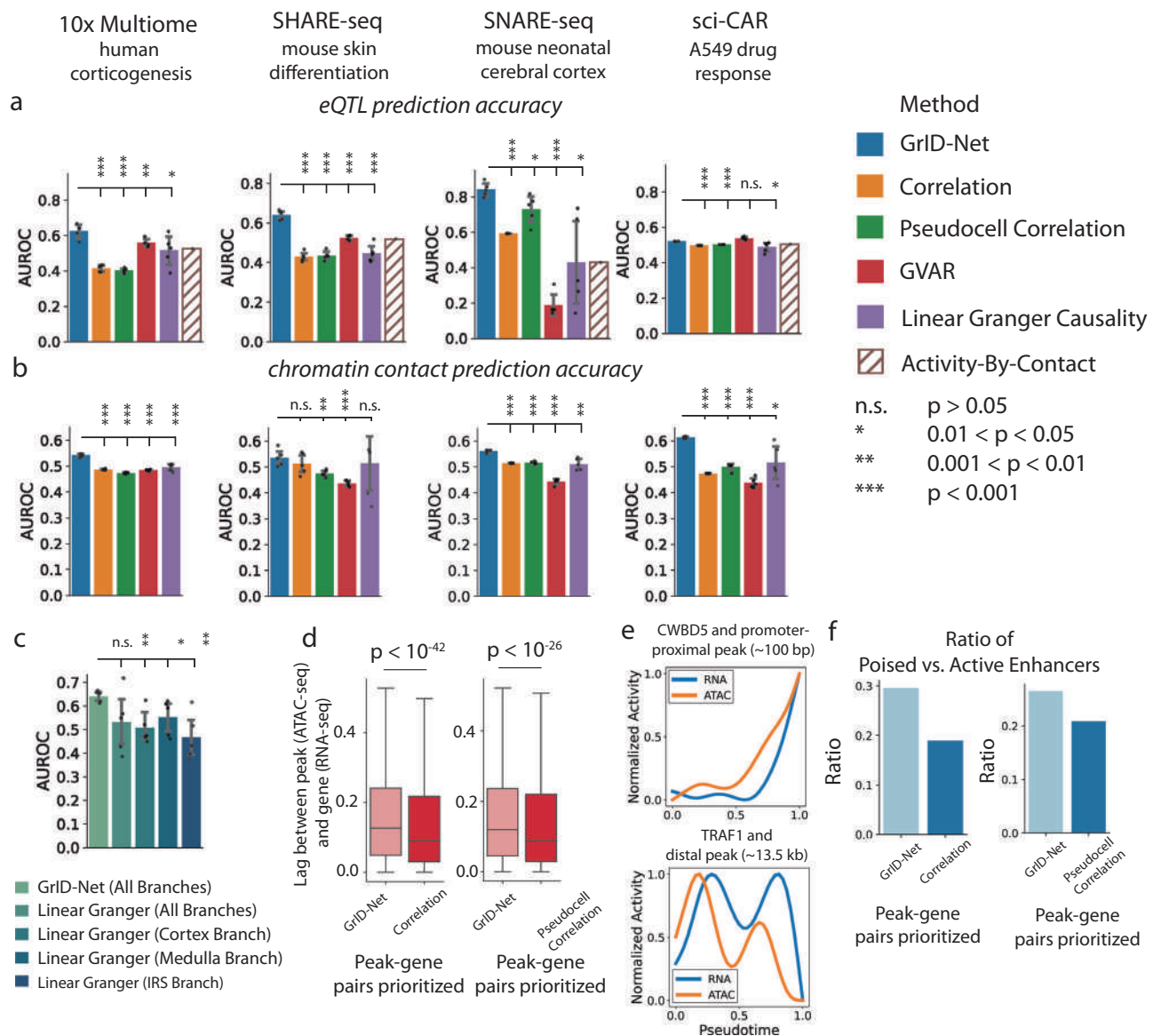


Fig. 2 | GrID-Net links noncoding loci to genes from single-cell multimodal dynamics. **a** Accuracy of single-cell (GrID-Net, Pearson correlation, and pseudocell correlation) and bulk (Activity-By-Contact) methods in predicting expression quantitative trait loci (eQTLs) in matching tissue across $n = 5$ uniformly sampled subsets of each of the single-cell multimodal datasets. Activity-by-Contact predictions were obtained from bulk data in matching tissue. We used eQTLs as proxies for causal interactions between genomic loci and genes. GrID-Net's out-performance of other methods was evaluated using Welch's one-tailed t -test. Data are presented as mean values $\pm$ SD (standard deviation). The same sampled subsets, statistical tests, and barplot data presentation are used in (b, c). **b** Accuracy of methods in predicting high-frequency three-dimensional chromatin contacts. The Activity-By-Contact model is excluded here, as it uses chromatin interactions data. **c** Accuracy of GrID-Net in predicting eQTLs in the SHARE-seq dataset compared to linear Granger causality when applied to all branches of the SHARE-seq trajectory and each of the branches separately. **d** Compared to existing approaches, GrID-Net is better able to identify asynchronous peak-gene associations. On the 10x

Multome dataset, we collected peak-gene pairs that GrID-Net prioritized differently than the correlation-based approaches, comparing the top 1% ($n = 14,802$) and bottom 1% ($n = 14,802$) of peak-gene pairs preferred by GrID-Net over each of the correlation and pseudocell correlation methods. For each pair, we quantified the asynchrony as the separation between pseudotime values corresponding to the cells where the peak and gene achieved their maximum values, respectively. The longer lags of peak-gene pairs prioritized by GrID-Net were evaluated using Welch's one-tailed t -test. The box in the box plot corresponds to the quartiles of the data, and the whiskers extend to the farthest data point within 1.5 times the interquartile range from the nearest hinge. **e** Lagged regulatory relationships predicted by GrID-Net between the expression of *CWBD5* and its promoter's accessibility, as well as between the expression of *TRAF1* and a distal peak 13.5 kb from its TSS. **f** GrID-Net effectively captures instances of chromatin priming. On the 10x Multiome dataset, we show the ratio of poised enhancers to active enhancers for peaks in peak-gene pairs prioritized by GrID-Net vs. each of the correlation-based methods. Source data are provided as a Source Data file.

identified longer-ranged lagged relationships, including a link between the expression of the neuroprotective gene *TRAF1*³⁴ and a peak 13.5 kb away from its TSS that has also been linked to *TRAF1* via eQTL analyses²⁸ (Fig. 2e).

Moreover, to explore the regulatory implications of this asynchrony, we obtained data on histone modification states of enhancers in neural progenitor cells and cross-referenced it with ATAC-seq

peaks³⁵. In peak-gene pairs differentially prioritized by GrID-Net, more poised enhancers (H3K4me1 only) overlapped with the peaks than active enhancers (both H3K4me1 and H3K27ac)³⁶ (Fig. 2f, Methods). This was not the case for peak-gene pairs differentially prioritized by correlation-based methods. Poised enhancers are regulatory sites that can trigger future gene expression changes but have not yet affected expression^{36,37}; in contrast, active enhancers correspond to ongoing

transcription³⁸. This suggests that resolving cell-state parallax enables the detection of lagged relationships between the activities of enhancers and their target genes. GrID-Net can thus leverage the concept of chromatin regulatory potential²⁶ to facilitate a deeper understanding of tissue-specific causal regulatory mechanisms.

To evaluate the biological significance of capturing these putative causal interactions, we examined the functions of genes implicated in peak–gene pairs differentially prioritized by GrID-Net. Genes in the peak–gene pairs that were specifically prioritized by GrID-Net over the correlation-based methods were highly enriched for gene sets associated with neurogenesis regulation and neural progenitor cells ($FDR < 10^{-7}$, GSEA Preranked³⁹), which are directly relating to the process of corticogenesis profiled in this dataset. In comparison, genes associated with peak–gene pairs that were not specific to GrID-Net were functionally enriched for housekeeping genes rather than neuronal activity. This disparity in enriched gene sets between the GrID-Net specific and non-specific peak–gene pairs suggests that GrID-Net's ability to detect poised enhancers is key to discovering cell type-specific regulatory relationships. Gene expression control during differentiation has been reported to depend highly on asynchronous enhancer–gene interactions⁴⁰. The enrichment of these differentiation-related gene sets thus underscores GrID-Net's ability to construct accurate genome-wide maps of noncoding locus–gene links that go beyond correlative analyses and towards causality.

Linking variants to schizophrenia disease genes

The ability to link noncoding loci to the genes they regulate enables the discovery of molecular mechanisms underlying complex diseases. Genome-wide association studies (GWAS) have identified thousands of loci associated with diseases⁴¹. However, over 90% of GWAS variants lie in noncoding regions⁴², and interpreting the functions of these noncoding variants involves the challenge of linking them to the specific genes that they regulate in a cell type-specific manner. With GrID-Net, we can leverage data from a single-cell assay on matching tissue to uncover these cell type-specific variant–gene links and help bridge the gap between genetic variants and gene dysregulation in disease.

We used GrID-Net to study variants associated with schizophrenia (SCZ), a psychiatric disorder for which inherited genetic variants are believed to underlie its pathogenesis⁴³. SCZ's high prevalence of around 1%, its substantial morbidity and mortality, and the lack of effective treatments are largely due to our limited understanding of SCZ pathophysiology^{44,45}. We applied the GrID-Net peak–gene links inferred from human corticogenesis²⁷, to interpret and analyze variants reported in the largest GWAS of SCZ to date⁴⁶. We refer to this original study as the Trubetsky et al. study.

To boost GrID-Net's accuracy on shorter-range peak–gene interactions, we incorporated one-dimensional genomic distance as an additional feature for predicting peak–gene associations. We observed a higher likelihood of peak–gene interaction if the two are located close to each other, in line with reported patterns of genomic distance-dependent regulatory effects^{47,48}. We calibrated the relative importance of this feature on a dataset profiling mouse skin differentiation²⁶ where we fitted a logistic regression model that combines the raw GrID-Net score with a genomic distance score (Methods). We found that integrating genomic distance with GrID-Net improved predictions of peak–gene links for mouse skin cells. The discriminative power on short-range peak–gene interactions improved, while the ability to detect long-range interactions was preserved. Furthermore, this model trained on mouse skin cell data performed well across species and cell types in human corticogenesis as well (Supplementary Fig. S6). This genomic distance-dependent model's scores were used for all remaining analyses.

From the Trubetsky et al. SCZ GWAS study, we retrieved all fine-mapped genetic variants with posterior probability (PP) over 5%. We intersected these variants with peaks from the 10x Multiome human

corticogenesis dataset. We then generated an initial set of candidate SCZ genes by associating each variant with up to 3 of its overlapping peak's highest scoring genes, retaining only variant–gene associations with scores in the top 50% of all peak–gene scores. This approach yielded a set of 132 genes that not only enriched for genes implicated in the original SCZ GWAS study ($FDR = 0.037$, Enrichr⁴⁹) but also included previously undetected SCZ gene candidates (Supplementary Data 1). For example, of the 12 SCZ variants that were prioritized by both GrID-Net and Trubetsky et al., 9 were linked to the same target genes by both approaches (Fig. 3a). This overlap included the association between rs324017 and *NAB2*, a gene implicated in stress-induced neuroprotective maintenance that also is the downstream target of the SCZ drug cinnarizine^{50,51}. For these same 12 SCZ variants, GrID-Net also proposed 20 additional SCZ candidate genes, of which 12 (60%) were found to be specifically expressed in brain tissues and thus potentially relevant in SCZ etiology (Methods). Some of these additional candidates include *NXPH4*, a neurexophilin involved in synapse function and cerebellar motor control⁵², and *EPN2*, an epsin linked to neural fate commitment⁵³. Moreover, of the 78 SCZ variants with PP over 50%, only 22 (28%) were linked to genes by the original SCZ GWAS study (Fig. 3b). Inclusion of SCZ variants that were linked to genes by GrID-Net increased this total to 30 (38%) SCZ variants. GrID-Net can, therefore, be used to complement standard strategies for linking variants to genes, such as those used in the original SCZ GWAS study. It can propose gene associations for variants with no annotations and also augment existing annotations by suggesting additional genes that a variant may be affecting.

Importantly, GrID-Net helps address a key challenge in GWAS analyses: the common strategy of assigning GWAS variants to their closest gene⁵⁴ misses many biologically meaningful variant–gene relationships. By leveraging single-cell heterogeneity, GrID-Net can evaluate longer-range interactions. Many of the variant–gene links we identify are characterized by large genomic distances between the variant and the TSS of the linked gene. Moreover, the variants are often linked to genes that are multiple genes over from their closest gene (Fig. 3c). For example, *DCPIB* and *NCKAPSL*, each of which are connected to only a single variant, are separated from their respective variants by 18 and 11 genes (235 and 242 kilobases (kb)), respectively. *DCPIB* has been implicated in SCZ, hypomania, and memory performance^{55,56}, while *NCKAPSL*, a cytoskeletal gene involved in neuronal activity⁵⁷, has previously been associated with SCZ as well as amyotrophic lateral sclerosis⁵⁸, indicating possible roles for these genes in neuron-specific dysregulation in SCZ.

In addition, 52 candidate genes (39.4% of the overall set) were associated with more than one variant (Fig. 3d). These genes may be controlled by the coordinated activity of multiple regulatory elements and co-regulators relevant to SCZ⁵⁹. *IGSF9B* was linked to 7 noncoding SCZ variants—the most of any gene. Coding variants in *IGSF9B* have been found to be associated with major depression and the negative symptoms of SCZ^{60,61}, underscoring its role in regulating affective symptoms in SCZ. Similarly, *KCNG2*, a potassium channel-encoding gene, was linked to 6 noncoding SCZ variants, and has been proposed as a drug target for SCZ, given the known disruption of potassium channel-related pathways in SCZ^{45,62,63}. Neither of these genes was prioritized in the original GWAS analysis.

We also assessed the cell type specificity of the genes linked to SCZ variants. We found that 104 candidate SCZ genes (78.8% of the overall set) are differentially expressed across the cell types profiled in the 10x Multiome human corticogenesis dataset (Welch's t -test with Benjamini-Hochberg correction $P < 0.05$, Fig. 3e). Further analysis of the SCZ variants also revealed that 59% of the variants lie in intronic regions, while only 7% of variants lie in intergenic regions more than 5 kb from any gene (Fig. 3f, Methods). As enhancers with tissue-specific effects are highly enriched in introns while intergenic enhancers more often regulate housekeeping genes⁶⁴, these findings support the

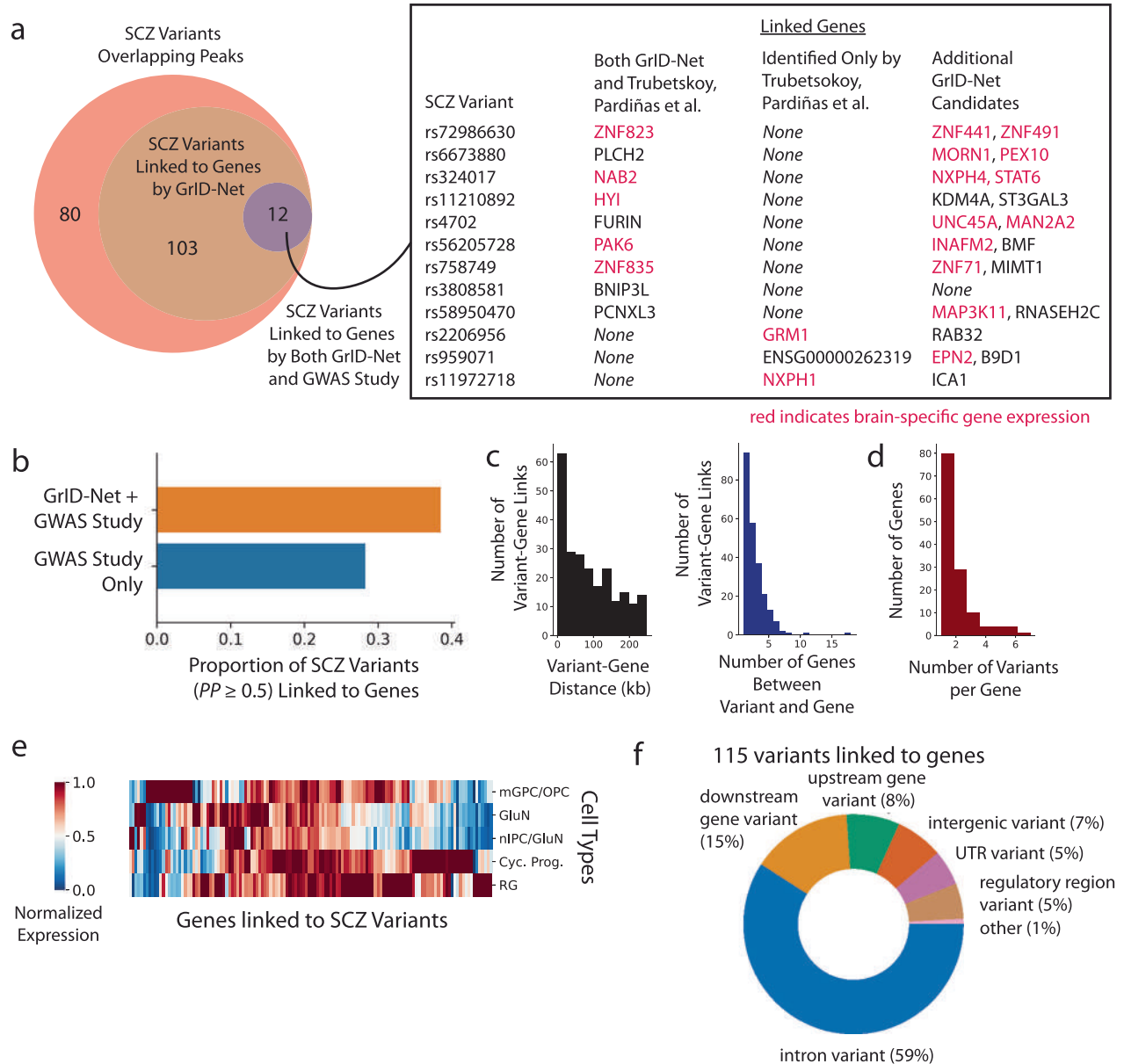


Fig. 3 | GrID-Net predictions link schizophrenia variants to affected genes.

a, b Schizophrenia (SCZ) variants from the Trubetsokoy et al. SCZ GWAS study prioritized by GrID-Net. Overall, 195 SCZ variants overlapped with scATAC-seq peaks from the human corticogenesis dataset. Compared to the original study, GrID-Net enables the annotation of many additional variants in this subset. Furthermore, as multimodal single-cell assays become common, we expect GrID-Net's coverage to widen. We also compared specific SCZ variant-gene links for the 12 variants where the original study also reported an annotation. **c** GrID-Net is able to identify long-range variant gene associations: (Left) genomic distance between an SCZ variant and the transcription start site (TSS) of its affected gene(s). (Right) number of genes in the genomic region between an SCZ variant and its affected

gene(s). **d** Histogram of the number of variants linked to each SCZ gene. **e** Genes linked to SCZ variants are cell type-specific. We show the heatmap of normalized expression values for all genes linked to SCZ variants in cell types involved in human corticogenesis. GluN glutamatergic neurons, nIPC neuronal intermediate progenitor cells, mGPC multipotent glial progenitor cells, OPC oligodendrocyte progenitor cells, Cyc. Prog. cycling progenitor cells, RG radial glial cells. A list of SCZ genes is included in Supplementary Data 1. **f** Distribution of annotated genomic locations for SCZ variants linked to genes. A substantial fraction of genes identified by GrID-Net are associated with intronic variants, which have been reported to have tissue-specific regulatory effects. Source data are provided as a Source Data file.

potential cell-type-specificity of the regulatory disruptions underlying SCZ pathogenesis.

Transcription factor binding disruptions are implicated in schizophrenia

Transcription factor (TF) binding at the cis-regulatory elements located in noncoding genomic loci is a key mechanism of gene regulation. We observed that SCZ variants with high PP are associated with larger TF binding affinity changes (Supplementary Fig. S7a). To investigate

the dysregulatory role of SCZ variants prioritized by GrID-Net, we, therefore, evaluated their effect on TF binding. For each variant associated with a gene by GrID-Net, we calculated the difference in the predicted TF binding affinity between the reference allele and the SCZ risk allele (Methods, Supplementary Data 2). Of the 115 SCZ variants linked to genes, 28 (these were associated with 49 target genes) were predicted to cause a 20-fold or stronger change in at least one TF's binding affinity (Fig. 4a). The TFs we identified as having the most widespread binding affinity changes are known to be implicated in

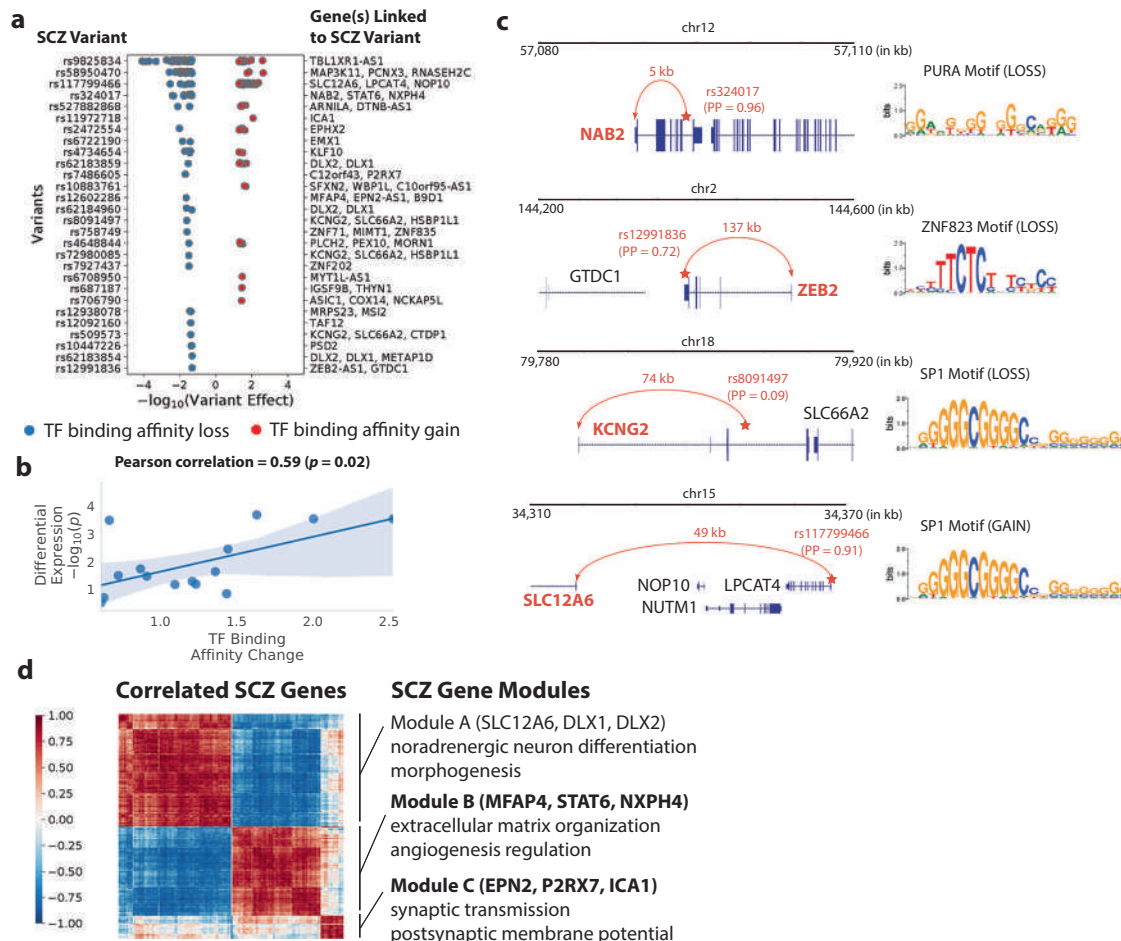


Fig. 4 | GrID-Net links transcription factor binding disruptions to gene dysregulation in schizophrenia. **a** TFs with at least a 20-fold change in predicted binding affinity caused by SCZ variants and the genes linked to these variants. Many of these TFs (e.g., KLF and SP families) are implicated in SCZ. **b** Relationship between TF binding affinity changes caused by SCZ variants and significance of differential expression in their linked genes between patient-derived SCZ glial

progenitor cells and matching controls. The significance of the relationship was evaluated via Pearson correlation analysis. **c** Examples of SCZ variant-gene links supported by at least 20-fold changes in predicted TF binding affinity. **d** Gene modules co-expressed with SCZ genes in SCZ GPCs and matching controls. Source data are provided as a Source Data file.

SCZ, exemplifying GrID-Net's ability to deliver de novo mechanistic insight. *VEZF1*, a vasculature-related TF reported to play a role in SCZ dysregulation⁶⁵, displayed diminished binding affinity most frequently (in five variants). In addition, *KLF15* and *KLF6*, two Kruppel-like family (KLF) TFs known to regulate neuronal growth and linked to SCZ^{46,66}, were each associated with reduced binding affinity in 4 variants (Supplementary Fig. S7b). Meanwhile, the TFs with the most widespread increase in binding affinity—*SPI*, *SP3*, and *KLF16*—have also been reported to be disrupted in SCZ^{67,68} (Supplementary Fig. S7c). More broadly, of the TFs with a 20-fold binding affinity change in at least one variant, 57.9% (=33/57) have cell type-specific expression patterns in corticogenesis (Welch's *t*-test with Benjamini-Hochberg correction $P < 0.05$). Collectively, these results point to genetic disruption of neuron-specific regulatory elements as potential drivers of SCZ pathogenesis.

We reasoned that if the TF binding affinity changes at a SCZ variant are phenotypically meaningful, they should result in expression changes of the corresponding gene(s) (as identified by GrID-Net). Accordingly, we obtained gene expression data from glial progenitor cells (GPCs) produced from SCZ patient-derived induced pluripotent cells, as well as matching controls^{63,69}, and computed differential expression patterns in SCZ GPCs. Focusing on glial differentiation genes, we observed that the differential expression of a gene in SCZ GPCs was indeed significantly correlated with TF binding affinity

changes at the corresponding SCZ variant (Pearson correlation coefficient = 0.59, Fig. 4b, Methods). The consistency between these independent observations provides support to our overall analytic approach and underscores the potential to leverage GrID-Net for examining causal, gene-specific regulatory mechanisms.

We drilled down into specific genes reported by our analysis, focusing on those linked to SCZ variants associated with at least 20-fold TF binding affinity changes (Fig. 4c). *NAB2*, the aforementioned gene implicated in neuroprotective processes⁵¹, was linked to rs324017, which contributes to a 200-fold decline in binding affinity for the TF *PURA*. Intriguingly, *PURA* is also critical for neural function and development, and its dysfunction is implicated in several neurological and cognitive impairments^{70,71}. In addition, rs12991836 was linked to *ZEB2*, whose TSS is positioned 137 kb from the variant. *ZEB2* has been previously reported as a target of long-range regulation, and heterozygous *ZEB2* mutations are known to cause Mowat-Wilson syndrome, which features structural brain abnormalities and intellectual disability⁷².

Furthermore, GrID-Net prioritized two potassium transport genes—*KCNG2* and *SLC12A6*—a class of genes which has been associated with compromised potassium uptake when dysregulated⁶³. This phenotype is characteristic of not only SCZ but also other neuron-related diseases, such as Huntington's disease, multiple sclerosis, and hemiplegic migraine^{69,73}. *KCNG2* was linked to 6 SCZ variants, the second most of

any gene, as described earlier. Using TF binding affinity changes to hone in on the most relevant variants, we identified rs8091497 as the variant with the largest TF binding affinity change. rs8091497 was associated with a 40-fold decline in binding affinity for *SP1*, which itself has been linked to behavioral abnormalities in SCZ⁶⁸. As for *SLC12A6*, its corresponding variant (rs117799466) increases the binding affinity for many TFs with repressive capabilities. The three TFs with the greatest binding affinity gains at this locus—*SP1*, *ZN281*, and *MAZ*—all play a role in neuronal differentiation^{68,74,75}, and both *SP1* and *ZN281* have also shown increased expression levels in SCZ as well^{76,77}. Combined with their increased binding affinity incurred by rs117799466, the upregulation of these TFs may contribute to the repression of *SLC12A6*, an outcome that has been previously linked to the potassium uptake impairments that characterize SCZ glia⁶³. Together, these findings suggest a possible mechanism for *SLC12A6* downregulation in SCZ that implicates enhanced binding of neuron-related repressors to a *SLC12A6*-targeting enhancer to reduce its expression levels.

We then broadened our analysis to gene modules, exploring if the genes identified by GrID-Net are systematically involved in SCZ-related pathways. We focused on the 14 most disruptive SCZ variants that cause at least 40-fold changes in TF binding affinity. These SCZ variants are linked by GrID-Net to 26 unique genes. To identify pathways that may be functionally impacted by disruptions of this set of genes, we searched for additional genes that are co-expressed with these genes in SCZ GPCs. In doing so, we detected 3 distinct gene modules that are highly co-expressed with several of the 26 GrID-Net-linked genes (Fig. 4d, Methods). Module A (*SLC12A6*, *DLX1*, and *DLX2*) is enriched for genes related to noradrenergic neuron differentiation ($FDR < 0.05$) and morphogenesis ($FDR < 0.02$); Module B (*MFAP4*, *STAT6*, *NXPH4*) is enriched for extracellular matrix organization ($FDR < 10^{-14}$) and angiogenesis regulation ($FDR < 0.02$) genes, and Module C (*EPN2*, *P2RX7*, *ICAI*) is enriched for synaptic transmission ($FDR < 10^{-4}$) and postsynaptic membrane potential ($FDR < 0.05$) (Enrichr⁴⁹). Each of these pathways has been previously associated with SCZ. Noradrenergic neurons have long been implicated in disrupted neuromodulation in SCZ⁷⁸, neuronal angiogenesis underlies the vascular abnormalities observed in SCZ⁷⁹, and genetic studies have pointed to the critical role of synaptic function in SCZ⁴⁶. These pathways, therefore, enable the interpretation of SCZ variants beyond the dysregulation of the specific genes that they directly affect, presenting an opportunity to discover systems-level mechanisms of SCZ pathogenesis.

Discussion

Most methods for single-cell multimodal data integration have focused on synchronizing modalities to improve downstream tasks, such as cell type identification. In contrast, we demonstrate it is also powerful to systematically study the asynchrony between the modalities. We presented GrID-Net, a method to uncover causal connections between genes and noncoding genomic loci using co-assayed single-cell RNA-seq and ATAC-seq data. GrID-Net operates at a genome-wide scale, is tissue-specific, and can effectively detect long-range (up to 1 Mb) cis-regulatory associations. A longstanding challenge with GWAS studies has been to identify the specific genes that are dysregulated by noncoding disease-linked variants. With GrID-Net, a single commercially available assay of the matching tissue can power a genome-wide estimation of variant-gene relationships. By resolving tissue-specific causal variant-gene associations, GrID-Net can help identify disease-linked genes as therapeutic targets.

When studying a particular disease, GrID-Net should be paired with a single-cell assay of a matching tissue undergoing a dynamic biological process, such as development, differentiation, disease, or drug response. GrID-Net can then translate the results of this single assay to genome-wide causal hypotheses specific to the disease. We demonstrate this by leveraging a single-cell multimodal assay of

human corticogenesis to interpret the functional effects of schizophrenia (SCZ) variants. GrID-Net expands both the breadth and depth of existing gene prioritization analyses: it finds associations for many unlinked variants while also uncovering additional causal mechanisms for previously investigated variants. For instance, we identified SCZ variants causing large (>20-fold) changes in TF binding affinity and connected them to 49 genes. Many of these genes are located distally from the SCZ variants that are predicted to affect them and would have been neglected by existing GWAS variant-gene associations analyses that consider only genomic proximity. Furthermore, by correlating TF binding affinity changes with differential expression in SCZ glial progenitor cells, we predicted possible mechanisms of TF-driven dysregulation for genes like *KCNQ2* and *SLC12A6*. Combined with an exploration of SCZ-related pathways associated with GrID-Net-identified genes, these analyses provide a comprehensive strategy for unveiling mechanistic insights about SCZ pathogenesis. Such analyses can also be applied to investigate other diseases.

Our work rests on two conceptual insights that are broadly applicable. First, when different modalities are measured in the same cell, their causal relationship implies a temporal offset between them, a phenomenon we refer to as “cell-state parallax”. Here, we exploit the parallax between RNA-seq and ATAC-seq, but such parallax can also be leveraged from other co-assayed modalities such as chromatin structure or methylation. We view RNA velocity⁸⁰ and protein velocity⁸¹ as specific applications of the broader concept of cell-state parallax, the distinction being that velocity uses pre-defined mappings between modalities (e.g., spliced vs. unspliced transcript-counts for a particular gene) to infer the differentiation direction of each cell. In contrast, such mappings between modalities may be unclear in general cell-state parallax and require customized inference approaches. Accordingly, our second conceptual insight is to develop computational frameworks that can accommodate the cross-modality dynamics of cell differentiation. Much of the existing work on modeling gene expression dynamics utilizes mathematical and statistical tools designed for time-series analysis^{82,83}, which make the unrealistic assumption that a total ordering of cells can capture the underlying biology. We argue that a partial ordering of cells is necessary to accurately model forks and merges in trajectories and, accordingly, formulate the differentiation landscape as a directed acyclic graph of cells that can represent such an ordering. We expect a promising direction of future research will be computational inference techniques, like GrID-Net here and the recently-introduced RNA velocity regulatory inference method Velorama⁸⁴, that can operate under this more realistic model.

Because GrID-Net operates on observational data, the locus-gene associations it infers are causal hypotheses that need to be validated by perturbations. In the eQTL-based evaluation, we sought to mimic such validations by investigating random genetic variants observed in the population and cross-referencing them to observed expression changes. We also note that such causal interactions are likely indirect and mediated by additional biophysical events, such as transcription factor binding. Nonetheless, indirect causality is biologically and therapeutically valuable, as the underlying mechanisms can still be exploited via interventions to achieve the desired phenotype.

While GrID-Net represents a significant advancement in causal inference for single-cell multimodal data, some limitations should be considered. One is the reliance on multiomic single-cell data from matching tissue undergoing a dynamic process, such as differentiation or disease, to provide the pseudotemporal context necessary for estimating chromatin accessibility and gene expression relationships. Although precise tissue matching is ideal, it is not strictly required to obtain meaningful results. For example, in our analysis of schizophrenia (SCZ) GWAS data, we leveraged corticogenesis data from the 10x Multiome study, which may explain only a subset of SCZ phenotypes. Nonetheless, the convenience and accessibility of the commercially available 10x Multiome assay allowed us to successfully infer

causal enhancer-gene links. Incorporating additional modalities, such as 3D genome structure from Hi-C assays or CTCF-based chromatin topology estimates, could refine predictions by better defining the genomic range within which enhancer-gene pairs are likely to interact. Another practical limitation is the computational burden imposed by large single-cell datasets, which may exceed GPU memory capacity. To mitigate this, we recommend using sketching techniques like Geosketch or Hopper, which have been shown to produce samples that retain the original dataset's biological heterogeneity^{85,86}. Since GrID-Net considers each cell as an independent datapoint, it is able to operate at a truly single-cell resolution, without needing to aggregate the data into pseudocells; we found that even the small sci-CAR dataset (~3000 cells) was sufficient to produce meaningful results. Additionally, if the goal is to focus on a specific phase of a biological process rather than the entire trajectory, subsampling can be particularly beneficial, allowing GrID-Net to operate efficiently while maintaining robust causal inferences.

With the growing availability of single-cell multimodal datasets, our approach is positioned to serve as a key framework for unlocking new mechanisms of gene regulatory dynamics that underlie fundamental biology and disease. Moreover, GrID-Net can be applied broadly to other domains featuring graph-based dynamics, such as uncovering the drivers of social media disinformation cascades. A key technical advance of this work is generalizing the applicability of standard Granger causal analysis to a wider set of domains than previously possible. Doing so required the use of graph-theoretic deep learning techniques to implement the core statistical intuition of Granger causality for any DAG-structured ordering of observations. Our work, like others before us⁸⁷, thus highlights how classical statistical concepts can be reimaged with modern deep learning techniques to dramatically widen their applicability.

Methods

GrID-Net formulation

See Supplementary Information.

GrID-Net training details and hyperparameters

All GrID-Net models consisted of $L = 10$ GNN layers and were trained using the Adam optimizer with a learning rate of 0.001 and a mini-batch size of 1024 candidate peak-gene pairs. Models were trained for a maximum of 20 epochs or until the relative change in the loss function was less than $0.1/|P|$ between successive epochs. Here, $|P|$ denotes the number of candidate peak-gene pairs evaluated per dataset. GrID-Net models were implemented in PyTorch and trained on a single NVIDIA Tesla V100 GPU.

Single-cell multimodal datasets

We used four single-cell multimodal datasets that spanned a range of different multimodal profiling techniques: (1) a 10x Multiome dataset profiling human corticogenesis, (2) a SHARE-seq dataset profiling mouse skin differentiation, (3) a SNARE-seq dataset profiling mouse neonatal cerebral cortical cells, and (4) a sci-CAR dataset profiling A549 cells treated with dexamethasone. For each of these datasets, we followed recommended practice⁸⁸ and applied $\log(1 + \text{CPM}/100)$ and $\log(1 + \text{CPM}/10)$ transformations to the RNA-seq and ATAC-seq counts data, respectively, where CPM indicates counts per million. We then filtered out all genes that were represented in fewer than 1% of cells in the dataset and all peaks that were represented in fewer than 0.1% of cells. We assembled a set of candidate peak-gene pairs by selecting all peaks with centers within 1 Mb from each gene's TSS. For the 10x Multiome dataset, we only considered peak-gene pairs that were separated by 250 kb to cohere with the genomic distance threshold used in the original study²⁷. To consider multiple samples for each dataset, we generated 5 uniformly sampled sets of 5000 cells for each of the 10x Multiome, SHARE-seq, and SNARE-seq datasets, while we

generated 5 uniformly sampled sets of 3000 cells for the sci-CAR dataset due its smaller quantity of cells (3081 cells). For the 10x Multiome dataset, we honed in on the branching dynamics of cycling glial progenitors, so we retained cells that were annotated as being glutamatergic neurons, neuronal intermediate progenitor cells, multipotent glial progenitor cells, oligodendrocyte progenitor cells, cycling progenitor cells, and radial glial cells.

Directed acyclic graph (DAG) construction

For each sample of a dataset, we first generated per-cell representations for each of the gene expression and chromatin accessibility modalities. Gene expression representations were constructed by (1) scaling the transformed RNA-seq data to unit variance and zero mean, (2) retaining the top 2000 most highly variable genes, (3) and projecting these normalized expression values onto their top 100 principal components (excluding components that have Spearman correlation $\rho > 0.9$ with the total counts per cell). Chromatin accessibility representations were obtained transforming the ATAC-seq data using tf-idf ^{89,90} and projecting the resulting values onto the top principal components as in (3) above. We then used Schema²⁰ to learn cell representations that unify information from these gene expression and chromatin accessibility representations. Specifically, we set the gene expression representation as the primary modality space in Schema and scaled each of the dimensions in this space such that the distances in the transformed space have a Pearson correlation coefficient of at least 0.99 with distances in the chromatin accessibility space. We constructed a k -nearest neighbor graph ($k = 15$) based on the Euclidean distances between these cell representations and inferred pseudotime using the scanpy⁹¹ implementation of diffusion pseudotime⁹² with default parameters. We then removed edges in the k -nearest neighbor graph if they did not align with the direction of increasing pseudotime, thus ensuring that no cycles exist and the resulting graph is a DAG.

Alternative methods

Pearson correlation. We computed the Pearson correlation between the transformed ATAC-seq and RNA-seq values for each candidate peak-gene pair, and the absolute value of the correlation coefficients was used to score the peak-gene pairs.

Pseudocell correlation. For each sample, we identified the 49 nearest neighbors for each cell based on the Euclidean distances between the joint representations learned from Schema (see “Directed acyclic graph construction”). We constructed a “pseudocell” from each cell and its 49 nearest neighbors, constructing its RNA-seq and ATAC-seq profiles by summing the raw counts from each of the modalities for all cells included in the pseudocell. We then applied $\log(1 + \text{CPM}/100)$ and $\log(1 + \text{CPM}/10)$ transformations to the RNA-seq and ATAC-seq counts data, respectively. Pearson correlation was applied to the transformed ATAC-seq and RNA-seq values for each candidate peak-gene pair, and the absolute value of the correlation coefficients was used to score the peak-gene pairs.

Linear Granger causality. We computed pseudotime points for each cell in a dataset using the same approach for pseudotime inference employed in DAG construction. We then sorted cells with respect to pseudotime and assigned them into 100 evenly sized bins. We averaged the gene expression values and chromatin accessibility values within each bin to yield a time series of length 100. We applied the statsmodels⁹³ implementation of Granger causality inference on this time series. For each peak-gene pair, we used a maximum lag of 10, selected the model lag using the Akaike information criterion, and used a likelihood ratio test to compute the statistical significance of Granger causality for the selected model. Peak-gene pairs were scored

and ranked with respect to the significance of their respective likelihood ratio test.

Generalized Vector Autoregression (GVAR). Similar to the linear Granger causality approach, we sorted and aggregated cells into 100 bins with respect to pseudotime. For each peak-gene pair, we applied GVAR with a single 8-dimensional hidden layer, a sparsity-inducing penalty $\lambda=1$, and a time-inducing penalty $\gamma=1$. Peak-gene pairs were scored and ranked with respect to the significance of the nonlinear Granger causal interactions inferred by GVAR.

Activity-by-contact (ABC) model. We obtained ABC prediction scores in cell types matching the cell types profiled in the single-cell multimodal studies. For each single-cell study, we intersected the candidate peak-gene pairs with the ABC enhancer-gene pairs, assigning each peak-gene pair the corresponding ABC score. If no ABC enhancer-gene pair overlap was identified, the peak-gene pair was assigned a value of 0, as the ABC prediction scores were available only for enhancer-gene pairs that exceeded the threshold of 0.015 defined in the original study¹⁰. For single-cell studies that profiled mouse cells, we mapped ABC enhancer-gene pairs in matching human cell types to the mouse genome due to the strong conservation of epigenomes between human and mouse^{94,95}.

eQTL evaluations

We obtained eQTL data from the eQTL Catalogue²⁸ and used eQTL data specific to the cell type profiled in each single-cell multimodal study. For mouse tissues, we mapped eQTL results for matching human tissues to the mouse genome using LiftOver⁹⁶, as human and mouse epigenomes show strong conservation^{94,95}.

We first intersected eQTL variant-gene pairs with peak-gene pairs from the single-cell data and assigned each remaining peak-gene pair the most statistically significant eQTL. We then binned these peak-gene pairs according to the genomic distance between each gene's TSS and each peak's center, creating a bin for each of the following ranges of genomic distances: [1 bp, 1 kb), [1 kb, 10 kb), [10 kb, 100 kb), [100 kb, 1 Mb). We identified all peak-gene pairs that overlapped significant eQTL variant-gene pairs ($P < 0.001$), and these peak-gene pairs served as the positive examples. To generate a distance-matched set of negative examples, we determined the number of positive examples in each genomic distance bin and selected an equal number of non-significant eQTL variant-gene pairs with the highest P values for that bin.

The compiled sets of peak-gene pairs were scored using GrID-Net and the other existing methods for each of the five samples associated with each dataset. Peak-gene pairs were ranked according to their scores, and an AUROC value was computed for each ranking. GrID-Net's outperformance of other methods with respect to AUROC values was evaluated using Welch's one-tailed t -test. Error bars in the accompanying bar plots correspond to bootstrapped 95% confidence intervals.

Chromatin interactions evaluations

We obtained Hi-C data for cell types matching the 10x Multiome, SNARE-seq, and sci-CAR datasets^{97–99}. We used 5C data for the SHARE-seq dataset¹⁰⁰ due to the unavailability of mouse skin Hi-C data. Chromatin interactions were mapped to a peak-gene pair by intersecting the genomic windows in the chromatin interactions data with the peak and the TSS of the gene. For Hi-C data, peak-gene pairs that were associated with the top 1% of chromatin contact frequencies were treated as positive examples, while the remaining peak-gene pairs were considered negative examples. For the 5C dataset, which evaluated the statistical significance of chromatin loops, peak-gene pairs that were associated with significant looping interactions ($P < 0.001$) were treated as positive examples, and the remaining peak-

gene pairs were considered negative examples. Similar to the eQTL evaluations above, peak-gene pairs were scored for each of the five samples associated with each dataset, and GrID-Net's outperformance with respect to AUROC values was compared with other methods using Welch's one-tailed t -test.

GrID-Net vs. correlation peak-gene prioritization

For the 10x Multiome human corticogenesis dataset, we averaged the rankings of all candidate peak-gene pairs across the five trials for each of the GrID-Net, Pearson correlation, and pseudocell correlation methods. We then calculated the difference between GrID-Net's average rankings for each peak-gene pair with those of the Pearson correlation and pseudocell correlation methods. The peak-gene pairs with the most positive differences represent the peak-gene pairs that GrID-Net prioritizes over the method against which GrID-Net is being compared. To evaluate enrichment of gene sets in peak-gene pairs prioritized by GrID-Net, genes were ranked with respect to these differences, and enrichment was computed using GSEA Preranked³⁹.

Enhancer states

We obtained H3K27ac and H3K4me1 ChIP-seq data for all samples of human neural progenitor cells and fetal neural cells in ChIP-Atlas³⁵. We identified statistically significant ChIP-seq peaks (MACS2 $Q < 10^{-10}$) that were positioned fully within ATAC-seq peaks from the 10x Multiome human corticogenesis dataset. Active enhancers were defined as ATAC-seq peaks containing overlapping H3K4me1 and H3K27ac ChIP-seq peaks, while poised enhancers were defined as ATAC-seq peaks containing H3K4me1 ChIP-seq peaks and no H3K27ac ChIP-seq peaks.

Genomic distance-dependent GrID-Net scores

We constructed a general model that integrated GrID-Net scores with genomic distance to account for known patterns of genomic distance-dependent transcription factor (TF) regulatory effects^{47,48}. The model assumes that TF regulatory effects exponentially decay with increasing genomic distance. To learn the decay rate, we trained a logistic regression model to predict peak-gene pairs overlapping significant skin eQTLs ($P < 10^{-3}$) for the SHARE-seq mouse skin dataset.

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_1 \log(s_i) + \beta_2 \hat{d}_i \quad (1)$$

Here, π_i is the probability of peak-gene pair i being a positive example, s_i corresponds to the mean rankings of the GrID-Net scores across trials for the peak-gene pair, and $\hat{d}_i = \max(\log_{10}(d_i) - 4, 0)$, where d_i is the genomic distance in base pairs (bp) between the center of the peak and the TSS of the gene in the peak-gene pair. This formulation assumes that only distal interactions at genomic distances greater than 10 kb are affected by the genome distance-dependent TF regulatory effect.

Positive and negative examples were chosen as described in "eQTL evaluations". We held out one-third of the examples as a test set, and then performed nested 3-fold cross-validation on the remaining examples, selecting the model that yielded the highest AUROC across the folds. This model increased the AUROC on the test set compared to the predictions from the averaged GrID-Net rankings (Fig. S3a). We applied this pre-trained model on the 10x Multiome human corticogenesis dataset and found that this model outperformed the baseline predictions from the averaged GrID-Net rankings (Fig. S3b). These results demonstrated the effectiveness of incorporating genomic distance as an additional feature alongside scores learned from the single-cell multimodal data.

Linking SCZ GWAS variants to genes

We obtained FINEMAP posterior probabilities (PP) calculated in the original Trubetskoy et al. SCZ GWAS study⁴⁶. We retained all fine-

mapped variants with $PP \geq 0.05$ that overlapped peaks from the 10x Multiome human corticogenesis dataset. We identified the peak that matched each SCZ variant and, using the genomic distance-dependent GrID-Net scores, we linked the variant to up to 3 of the peak's highest-scoring genes, retaining only peak-gene pairs in the top 50% of all peak-gene scores.

Brain-specific gene expression

We obtained consensus tissue-specific transcript levels from the Human Protein Atlas^{101,102}. For a given gene, we first calculated the mean and standard deviation of the gene's transcript levels across all non-brain tissues. The brain-specific tissues were identified to be those from the basal ganglia, cerebellum, cerebral cortex, medulla oblongata, midbrain, pons, and white matter. We then converted the transcript level for the gene in each of the brain-specific tissues into a z-score with respect to the non-brain tissues ($z = \frac{x - \mu_{\text{non-brain}}}{\sigma_{\text{non-brain}}}$, where x is the transcript level for the gene in a brain-specific tissue). Genes that had a z-score over 1 in any of the brain-specific tissues were annotated as having brain-specific gene expression.

TF binding affinity changes

We obtained the reference and SCZ risk alleles for all SCZ variants from the original SCZ GWAS study⁴⁶. We then flanked each of the alleles with the genomic sequences ± 20 bp of the SCZ variant position using the GRCh38/hg38 reference genome¹⁰³, generating a reference sequence and SCZ variant sequence. We applied FIMO¹⁰⁴ to each of these sequences to predict TF binding affinities using position weight matrices (PWMs) from HOCOMOCO v11 (Full Collection)¹⁰⁵, JASPAR CORE (Non-Redundant) v2¹⁰⁶, and CIS-BP v2¹⁰⁷. We evaluated the effect of a specific SCZ variant on a TF's binding affinity by comparing the FIMO-calculated p values for the TF between the reference sequence and SCZ variant sequence.

SCZ GPC gene expression

We obtained bulk RNA-seq data from glial progenitor cells (GPCs) produced from four SCZ patient-derived induced pluripotent cells (iPSCs) and paired control subjects^{63,69}, with at least three samples per patient. We applied a $\log(1 + \text{CPM})$ transformation to this data.

To evaluate the relationship between TF binding affinity changes caused by SCZ variants to expression changes in GPCs, we first determined peak-gene pairs that are specific to glial differentiation. In the 10x Multiome human corticogenesis dataset, we identified genes that were differentially expressed between glial cells and the cycling progenitor cells, which were the root of the corticogenesis trajectory ($P < 0.05$, Welch's t -test). We then used Welch's t -test to compare the expression levels of these glial differentiation-specific genes for each SCZ patient against all of the control samples. For each gene, we identified the most statistically significant difference so as to account for all genes that may be differentially expressed in any of the patients compared to the controls.

SCZ GPC co-expression pathways

We first identified a set of 26 genes linked to SCZ variants causing at least a 40-fold change in binding affinity for any TF. We identified three sets of these SCZ genes that were correlated with each other (Pearson correlation coefficient ≥ 0.6) in the SCZ GPCs and paired controls (*SLC12A6*, *DLX1*, *DLX2*; *MFAP4*, *STAT6*, *NXPH4*; *EPN2*, *P2RX7*, *ICAI*). For each of these sets of SCZ genes, we then determined gene modules that may be implicated in their dysregulation by identifying genes that are highly co-expressed (Pearson correlation coefficient ≥ 0.8) with any of the SCZ genes in the set. For each of these three modules of highly co-expressed genes, we applied gene set enrichment analysis using Enrichr to detect biological processes that were enriched within its respective gene sets.

Statistics and reproducibility

No statistical method was used to predetermine sample size. As described in detail in "Single-cell multimodal datasets", subsets of cells were uniformly sampled from single-cell multimodal datasets when inferring peak-gene links, and our analysis of the 10x Multiome dataset focused on cells related to the branching dynamics of cycling glial progenitors. No other data were excluded from the analyses. The Investigators were not blinded to allocation during experiments and outcome assessment.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Code availability

The code used to develop the model, perform the analyses, and generate results in this study is publicly available and has been deposited at <https://github.com/alexw16/gridnet> under the MIT license. The specific version of the code associated with this publication is archived in Zenodo and is accessible via 10.5281/zenodo.1524981¹⁰⁸.

Data availability

All of the data included in this study are publicly available. The sci-CAR, SNARE-seq, SHARE-seq, and 10x Multiome datasets were downloaded from the Gene Expression Omnibus (GEO) database using the accession numbers GSE117089, GSE126074, GSE140203, and GSE162170, respectively. eQTL data were downloaded from the eQTL Catalogue [<https://www.ebi.ac.uk/eqtl>]. Hi-C and 5C datasets were obtained from the GEO database using the accession GSE77565, GSE35156, and GSE102519, as well as the ENCODE database [<https://www.encodeproject.org/>]. ChIP-seq data were downloaded from ChIP-Atlas [<https://chip-atlas.org/>]. Source data are provided with this paper.

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Author contributions

A.W., R.S. and B.B. designed research; A.W., R.S. and B.B. performed research; A.W. and R.S. analyzed data; and A.W., R.S., C.A.W. and B.B. wrote the paper.

Competing interests

Since September 2023, A.W. has been an employee of Genentech and has equity in Roche. The other authors have no competing interests.

Additional information

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